

ATHENA ITD

AI-Driven Insider Threat Detection Platform

User Guide & Complete Documentation

Version 3.2.0 · July 25, 2026

CLI · Web SOC Console · REST API · ML Detection Engine

Author: Petro Plaku · pplaku799@gmail.com

© 2026 Petro Plaku — All rights reserved.

Contents

…

> ℹ️ **Updating this table**
>
> In Word/LibreOffice: right-click the table of contents and choose **Update Field** (or press F9 / Ctrl+A then F9) to populate page numbers.

1. Welcome to ATHENA ITD

ATHENA ITD is an **AI-driven insider threat detection platform**. It ingests activity logs from your environment (authentication, file access, email, USB, cloud, database and network events), builds a behavioral baseline for every user, and combines a machine-learning anomaly ensemble with 18 explainable detectors to surface the users who are behaving like insider threats — data theft, sabotage, account compromise, privilege abuse — **before** the damage is done.

Everything ships in a single package with three equal interfaces:

Interface	What it is	Typical user
athena CLI	Full-featured terminal client — ingest, train, analyze, triage, respond, report	Analysts, automation, air-gapped hosts
Web SOC Console	13-view browser dashboard with live stream, cases, MITRE map and admin panels	SOC teams, managers
REST API	Every console capability as JSON endpoints, key- or session-authenticated	SIEM/SOAR engineers

1.1 What makes it different

- **Explainable by design** — every alert lists weighted, human-readable factors ("Moved 1.7 GB through exfiltration channels, 3.2σ above their normal"), never a bare score.
- **Hybrid detection** — an Isolation-Forest ML ensemble (30% weight) + rule detectors (45%) + peer-group deviation (15%) + rare-sequence analysis (10%) are fused into one 0–100 risk score.
- **MITRE ATT&CK & compliance native** — every detector maps to ATT&CK techniques and to GDPR, HIPAA, ISO 27001, SOC 2, PCI DSS, NIST 800-53 and NIST CSF controls.
- **Complete SOC workflow** — triage statuses, case management, watchlists, analyst feedback that measures per-detector precision, full audit trail.
- **Response, gated safely** — 8 response actions are simulated by default; real enforcement needs an explicit double opt-in.
- **12 forwarding integrations** — Splunk, Elastic, Microsoft Sentinel, generic webhook, Slack, Teams, PagerDuty, SMTP e-mail and syslog (CEF/LEEF/RFC 5424).

1.2 How detection works (30-second version)

Events → per-user-per-day **feature vectors** (36 features) → the ML ensemble scores each vector against the organization baseline → 18 detectors add weighted findings → scores are fused, watchlist multipliers applied, and each user receives a **0–100 risk score** mapped to LOW / MEDIUM / HIGH / CRITICAL. Alerts carry their full explanation, MITRE techniques and recommended responses.

Risk level	Default score band	Meaning

LOW	0 – 49	Normal behavioral envelope
MEDIUM	50 – 74	Unusual — review when convenient
HIGH	75 – 87	Investigate today
CRITICAL	88 – 100	Act now — likely active insider activity

2. Installation

2.1 Requirements

- Python **3.10+** (3.12 recommended) — or Docker
- ≈ 300 MB disk for the application + your event data (SQLite)
- Linux, macOS or Windows

2.2 Install with pip (recommended for a workstation)

```
# in a virtual environment
python3 -m venv athena && source athena/bin/activate
pip install athena-itd          # or: pip install ./athena_itd-3.2.0-py3-none-any.whl

athena version                  # verify: shows v3.2.0 and data paths
athena init                   # create config + database
```

2.3 Run with Docker / docker compose (recommended for a server)

```
git clone <your-repo>/athena-itd && cd athena-itd
echo "ATHENA_ADMIN_PASSWORD=$(openssl rand -base64 18)" > .env
echo "ATHENA_API_KEY=$(openssl rand -hex 32)" >> .env
docker compose up -d --build
curl -fsS http://localhost:8000/api/health   # → {"status":"ok"}
```

The container runs as a **non-root** user, stores all data on the **/data** volume, and has a built-in healthcheck. Put a TLS reverse proxy (Caddy file included in **deploy/**, or nginx/Traefik) in front for anything shared.

2.4 Where your data lives

Item	Default location	Override
Config	**<data-home>/config.json**	**ATHENA_HOME** env var
Database	**<data-home>/athena.sqlite3**	—
Trained model	**<data-home>/model.joblib**	—
Reports	**<data-home>/reports/**	—

<data-home> is the OS application-data directory (e.g. **~/.local/share/athena-itd** on Linux) unless **ATHENA_HOME** is set. **athena version** always prints the active paths.

3. Five-Minute Quick Start

The fastest way to see ATHENA work is the built-in synthetic dataset — it plants real insider-threat scenarios (data exfiltration, failed-login bursts, off-hours access) so you can watch them get caught.

Complete first run

```
athena init                       # 1. create config + DB
athena generate --users 25 --days 30 # 2. synthetic corpus with planted threats
athena train                      # 3. fit the ML ensemble
athena analyze                    # 4. score every user
athena alerts                     # 5. see who was caught
athena show <username>                  # 6. full explainable finding
athena serve                      # 7. open the web console
```

✓ First login to the console

On first start with no accounts, ATHENA creates an **admin** user. If **ATHENA_ADMIN_PASSWORD** is not set, a **random one-time password is generated and printed once in the server log** — copy it, sign in, then change it under Settings → User Accounts (or **athena users passwd admin**).

When you are done exploring, wipe the demo corpus with **athena reset** and ingest your real logs (next chapter).

4. Getting Your Data In

ATHENA normalizes everything into a single **Event** shape. Three file formats are auto-detected by **athena ingest <file>** (or the console's **Data → Upload** panel, or **POST /api/ingest**):

Format	Detection	Typical source
CSV	header row with **timestamp,user,action,…**	SIEM exports, HR exports, custom scripts
JSON lines	one JSON object per line	Cloud audit logs, custom pipelines
Auth log (syslog)	**sshd/sudo** syslog lines	Linux **/var/log/auth.log**

4.1 The CSV schema

Only **timestamp, user, action** are required. Every additional column unlocks more detectors:

Column	Required	Example	Used by
timestamp	yes	2026-02-01T09:02:11	everything (ISO-8601 or common formats)
user	yes	alice.wong	everything
action	yes	file_download	everything (see action list below)
host	no	WS-2201	lateral movement, remote sessions
source_ip	no	10.0.1.15	geo/anonymous-origin, IP intel
resource	no	/finance/q1.xlsx	sensitive-data detector, investigation detail
bytes	no	2400000	exfiltration volume, mass collection
success	no	true / false	failed-login burst
department	no	finance	peer-group deviation
role	no	analyst	peer-group deviation
country	no	US	impossible travel / new geography

4.2 Recognized actions

The **action** column accepts these values (anything else becomes **other**):

```
login  logout  failed_login  mfa_challenge  mfa_denied  password_change
vpn_connect  file_access  file_download  file_upload  file_delete
file_rename  file_encrypt  data_transfer  usb_insert  usb_write  print
cloud_upload  repo_clone  db_query  email_send  email_forward_rule
email_external  privilege_escalation  config_change  account_create
account_modify  account_delete  permission_grant  security_tool_disable
log_clear  remote_session  process_exec  web_browse  other
```

4.3 Ingest commands

```
athena ingest export.csv              # append to the corpus
athena ingest export.csv --reset      # replace the corpus first
athena ingest app.jsonl --format json  # force a format
athena ingest /var/log/auth.log       # sshd/sudo lines -> events
```

> ℹ️ **Mixing sources is encouraged**
>
> Ingest CSV exports, JSON audit trails and auth logs into the same corpus — ATHENA correlates them per user. Rows that cannot be parsed are skipped and counted, never fatal.

4.4 The CERT insider-threat dataset & synthetic data

```
athena dataset list                   # available loaders
athena dataset load cert --path ~/r4.2 # CMU CERT r4.2/r6.2 directory
athena dataset inspect cert --path ~/r4.2
athena generate --users 40 --days 60 --seed 7   # synthetic, reproducible
```

The CERT loader reads **logon.csv**, **device.csv**, **file.csv**, **email.csv** and the **LDAP/** directory from the public CMU CERT corpus — the standard academic benchmark for insider-threat research.

5. The Detection Engine

5.1 The 18 detectors

Key	Detector	What it catches
ml	ML behavioral anomaly	Isolation-Forest ensemble scores each user-day feature vector (36 features) against the organization baseline
exfil	Data exfiltration volume	Outbound volume z-score vs the user's own baseline; flags large transfers
usb	Removable-media exfiltration	USB insert + write bursts, volume moved to removable media
cloud	Personal cloud upload	Uploads to personal cloud storage services
email	Suspicious email egress	External mail volume, forward-rule creation, attachment spikes
failed	Failed-login burst	≥ 5 failures in a window (brute force / password guessing)
offhours	Off-hours access	Activity outside 08:00–19:00 business hours, weekends
privesc	Privilege escalation & abuse	Escalations, account/permission changes, admin-tool abuse
geo	Impossible travel / new geography	Logins from new countries or physically impossible sequences
anonip	Anonymised network origin	Tor exit nodes, VPN and hosting-provider source addresses
masscollect	Mass data collection	≥ 120 distinct files touched in a day (staging behavior)
massdelete	Mass deletion / destruction	≥ 40 deletions in a day (sabotage)
evasion	Defense evasion	Security-tool disabling, log clearing
lateral	Lateral movement	≥ 6 distinct hosts touched in a day
dormant	Dormant account activation	Accounts idle ≥ 21 days that suddenly act
sensitive	Sensitive-data access spike	Unusual access to resources matching sensitive patterns
peer	Peer-group deviation	2.5σ+ deviation from the user's department/role peer group
sequence	Rare action sequence	Action n-grams rarer than 1.5% of all observed sequences

5.2 Training the ML model

```
athena train        # fits the ensemble on the stored corpus, saves model.joblib
```

Training is optional but recommended: without a persisted model, **analyze** fits a fresh model on the fly each run. The trained ensemble is also what the **live stream** uses to

score events in real time. Re-train whenever you ingest a meaningfully larger or newer corpus.

5.3 Tuning: thresholds and detector weights

Everything is tunable — from the console (**Detections** view and **Settings**), the API (**POST /api/config**, **POST /api/detections**) or by editing **config.json**:

Threshold	Default	Meaning
low / medium / high / critical	25 / 50 / 75 / 88	Score → risk-level band edges
contamination	0.08	Expected anomaly fraction for the Isolation Forest
business_start/end_hour	8 / 19	Off-hours window boundaries
failed_login_burst	5	Failures per window to trigger **failed**
data_exfil_zscore	3.0	σ above personal baseline to trigger **exfil**
peer_deviation_sigma	2.5	σ from peer group to trigger **peer**
mass_access_files	120	Files/day for **masscollect**
mass_delete_files	40	Deletes/day for **massdelete**
lateral_host_count	6	Hosts/day for **lateral**
dormancy_days	21	Idle days for **dormant**
sequence_rarity	0.015	Rarity cut-off for **sequence**
new_geo_penalty	15.0	Points added for a new country
weight_ml / rules / peer / sequence	0.30 / 0.45 / 0.15 / 0.10	Fusion weights of the four scoring pillars
risk_halflife_days	7.0	Risk decay half-life

Individual detectors can be **disabled** or **re-weighted** (0.1×–3.0×) in the Detections view — useful to silence a noisy detector for your environment without losing the rest.

5.4 The analyst feedback loop

Every alert accepts a verdict — **true_positive**, **false_positive**, **benign_true_positive** or **unknown**. Verdicts feed a per-detector **precision score** (**athena feedback-stats** or the Detections view), showing exactly which detectors earn their keep in your environment. Marking an alert false-positive also moves it out of the active queue.

6. The Web SOC Console — View by View

```
athena serve                       # http://127.0.0.1:8000
athena serve --host 0.0.0.0 --port 8443 --tls-cert cert.pem --tls-key key.pem
athena serve --workers 4 --proxy-headers --access-log   # behind a proxy
athena gui                         # serve + open a browser
```

The console is a single-page application organized into four navigation groups and thirteen views:

Governance group

View	What you see & do
Overview	KPI tiles (events, users, analyzed, flagged), threat level, 30-day activity & byte trends, top risky users, model status, live-stream status
MITRE ATT&CK	Coverage matrix of all detector techniques, heat-mapped by live alert hits; per-technique detail
Compliance	Control coverage for GDPR, HIPAA, ISO 27001, SOC 2, PCI DSS, NIST 800-53, NIST CSF — with per-control evidence counts from actual alerts

Investigation group

View	What you see & do
Users	Full directory with risk score, level, department, top factor; search, department & min-level filters; click through to the investigation drawer
User drawer	Raw event tail, hour-of-day histogram, action breakdown, 30-day event/byte charts, alert explanation with MITRE detail, notes, watchlist toggle
Alerts	Triage queue: filter by level/status/user, set status (new → triaged → investigating → escalated → contained → resolved / false_positive / suppressed), assign owners, record verdicts, create cases
Cases	Investigation cases with severity, owner, linked alerts, notes timeline, HR/Legal pending states, closing verdicts
Geo Intel	Country activity map points, source-IP reputation categories (Tor / VPN / hosting / corporate), flagged anonymised origins, per-IP lookup

Operations group

View	What you see & do
Live	Real-time event stream (SSE) with on-the-fly

	scoring — start/stop the demo streamer, watch alerts fire as events arrive
Responses	Response-action catalog, dispatch actions per user, auto-respond from an alert's recommendations, full response audit trail
Data	Upload log files, load datasets (CERT/synthetic), generate demo data, reset the corpus, retention purge

Platform group

View	What you see & do
Detections	Enable/disable and weight all 18 detectors, edit every threshold, view per-detector precision from analyst feedback
Audit	Complete operator audit trail — who did what, to which target, from which IP, when
Settings	Organization name, thresholds, integrations (Splunk/Elastic/Sentinel/Slack/Teams/PagerDuty/syslog/SMTP/webhook), security (auth, sign-up, rate limits, CORS), retention policy, **User Accounts** admin, **Active Sessions** admin, API key

✓ Everything in the GUI is also an API call

The console consumes the same REST API documented in chapter 11 — anything you can click can be scripted.

7. Daily SOC Workflow

7.1 Triage an alert

- **1. Review** — open Alerts, sort by risk; read the weighted factors (each explains its evidence and σ-deviation).
- **2. Classify** — set a verdict: **true_positive**, **benign_true_positive** (real but authorized, e.g. a sanctioned bulk export), **false_positive**, or leave **unknown**.
- **3. Update status** — **triaged** → **investigating**; assign an owner.
- **4. Escalate** — create a **case** directly from the alert; the alert links automatically and moves to **investigating**.
- **5. Respond** — dispatch response actions (chapter 8) if warranted.
- **6. Close** — **resolved**, **contained** or **false_positive**.

7.2 Cases

```
athena cases list
athena cases open --title "Exfil – carol.diaz" --user carol.diaz --severity HIGH
athena cases close 3 --verdict true_positive
```

Cases carry severity, owner, tags, linked alerts, notes and a status lifecycle that includes **pending_hr** and **pending_legal** — because insider investigations are cross-functional. Closing states record whether the case was **closed_confirmed** or **closed_unsubstantiated**.

7.3 Watchlist

```
athena watchlist add jsmith --reason "resignation notice 2026-07-01" --multiplier 1.5 --days 30
athena watchlist list
athena watchlist remove jsmith
```

Watchlisted users get a **risk multiplier** (1.0×–3.0×) applied at scoring time, with optional expiry — the standard play for leavers, PIP cases and users under investigation.

8. Response Actions (SOAR-lite)

Action	Effect
lock_account	Lock/disable the user's account
revoke_sessions	Terminate all active sessions
require_mfa	Force MFA step-up at next access
disable_downloads	Block outbound file transfers / USB
isolate_endpoint	Quarantine the user's host from the network
revoke_privileges	Strip elevated privileges
notify_soc	Notify the SOC channel
notify_manager	Notify the user's manager

✋ **The enforcement gate — read before enabling**

All actions are **simulated by default**: they are recorded, audited and forwarded, but nothing real is executed. Real enforcement requires BOTH **security.allow_enforced_response = true** in the config AND **--force** / **"force": true** on the individual call. **athena doctor** warns whenever enforcement is enabled.

```
athena respond carol.diaz lock_account --reason "case #3"
curl -X POST https://athena/api/respond -H "X-API-Key: $KEY" \
    -d '{"user":"carol.diaz","action":"revoke_sessions","reason":"case #3"}'
```

Auto-respond executes the recommended action set attached to an alert (**athena respond --auto <user>** or **POST /api/respond/auto/{user}**). Every dispatch — simulated or real — lands in the response audit trail.

9. Alert Forwarding & Integrations

Alerts at or above the configured forwarding level are pushed to every configured destination when an analysis runs. Configure under **Settings → Integrations**, via env vars, or **POST /api/config**.

Destination	Config keys	Notes
Splunk	**splunk_hec_url**, **splunk_hec_token**	HTTP Event Collector JSON
Elastic	**elastic_url**, **elastic_api_key**, **elastic_index**	Index API (default index **athena-alerts**)
MS Sentinel	**sentinel_workspace_id**, **sentinel_shared_key**	Log Analytics HTTP Data Collector
Generic webhook	**webhook_url**	POST of the full alert JSON
Slack	**slack_webhook**	Rich message with factors
Microsoft Teams	**teams_webhook**	Adaptive card
PagerDuty	**pagerduty_routing_key**	Events API v2 — pages on-call
E-mail (SMTP)	**smtp_host/port/user/password/from/to**	Plain-text alert mail
Syslog	**syslog_host**, **syslog_port**, **syslog_format**	**CEF**, **LEEF** or **RFC 5424** — any SIEM

```
# example: forward HIGH+ alerts to Slack and Splunk
export ATHENA_SLACK_WEBHOOK=https://hooks.slack.com/services/T000/B000/xxx
export ATHENA_SPLUNK_HEC_URL=https://splunk:8088/services/collector
export ATHENA_SPLUNK_HEC_TOKEN=…
athena analyze        # delivery results are reported per destination
```

athena intel <feed.csv> additionally layers an operator-supplied IP reputation feed (ip,category rows) over the built-in Tor/VPN/hosting classifier used by the **anonip** and geo detectors.

10. Accounts, Roles & Security

10.1 Roles (RBAC)

Role	Can	Cannot
viewer	Read every dashboard, alert, case, report	Change anything
analyst	Everything viewer can + triage alerts, manage cases/notes/watchlist, dispatch responses	Platform config, accounts, data management, training
admin	Everything — config, detectors, accounts, sessions, ingest/reset, training, purge	—

10.2 Managing accounts

```
athena users list
athena users add jdoe --role analyst       # prompts for password (min 8 chars)
athena users passwd jdoe                   # rotate (revokes their sessions)
athena users role jdoe admin
athena users disable jdoe                  # blocks login + kills sessions
athena users sessions                      # live sign-ins
athena users revoke jdoe                   # sign a user out everywhere
```

The console equivalents live in **Settings → User Accounts / Active Sessions**. Self-service sign-up (**/signup**) is on by default, always creates non-admin accounts, is rate-limited per IP, and can be disabled with **ATHENA_ALLOW_SIGNUP=false**.

10.3 How ATHENA protects itself

- **Passwords**: scrypt (memory-hard) with per-user salt; 8-char minimum policy.
- **Sessions**: 32-byte random tokens; only the SHA-256 hash is stored; DB-backed with TTL (default 8 h), revocable, listed in the console.
- **First admin**: generated one-time password — no shared default credential, ever.
- **CSRF**: double-submit cookie required on every session-authenticated mutation.
- **Login throttling**: per user+IP exponential lockout after 5 failures.
- **Rate limiting**: per-caller request cap on the API; sign-up capped per IP.
- **Last-admin guards**: the final active admin cannot be deleted, demoted or deactivated.
- **Headers**: nosniff, DENY framing, no-referrer, HSTS when served over TLS.
- **Audit**: every operator action recorded with actor, target, IP and timestamp.
- **API key**: optional **X-API-Key** machine principal (admin-level, skips CSRF) for SIEM/SOAR automation — generate with **openssl rand -hex 32**.

> ⚠️ **Production checklist**
>
> Serve over **TLS**, set a strong **ATHENA_ADMIN_PASSWORD** before first start, decide on sign-up, set a retention policy, and run **athena doctor** — it grades your deployment posture and fails on lockout risks. See SECURITY.md for the full checklist.

11. REST API Reference

Authenticate with a session cookie (browser) or the **X-API-Key** header (automation). Roles: **V**=viewer, **A**=analyst, **Ad**=admin, **P**=public. Session mutations also need the **X-CSRF-Token** header.

Method & path	Role	Purpose
GET **/api/health** · **/api/ready** · **/api/metrics** · **/api/version**	P	Liveness, readiness, Prometheus metrics, build info
POST **/api/login** · **/api/logout** · **/api/signup**, GET **/api/me**	P	Auth flow
GET **/api/summary**	V	Dashboard KPIs, trends, threat level
GET **/api/users** · **/api/user/{u}** · **/api/departments**	V	Directory, investigation detail
GET **/api/alerts**, POST **/api/alert/{k}/status** · **/{k}/feedback**	V / A	Triage queue, status, verdicts
GET/POST **/api/cases**, GET/PATCH **/api/case/{id}**, POST **/api/case/{id}/link**	V / A	Case management
GET/POST **/api/notes/{type}/{id}**	V / A	Notes on alerts, cases, users
GET/POST **/api/watchlist**, DELETE **/api/watchlist/{u}**	V / A	Watchlist
GET **/api/mitre** · **/api/compliance** · **/api/geo** · **/api/ip/{ip}**	V	ATT&CK matrix, control coverage, geo intel
GET **/api/audit** · **/api/export** · **/api/export.csv** · **/api/responses**	V	Audit trail, exports, response log
POST **/api/respond** · **/api/respond/auto/{u}**, GET **/api/response/actions**	A	Dispatch response actions
POST **/api/analyze**	A	Run detection over the corpus
GET/POST **/api/detections** · **/api/config**	V·Ad / Ad	Detector & platform configuration
POST **/api/train** · **/api/generate** · **/api/ingest** · **/api/dataset/load** · **/api/reset** · **/api/purge**	Ad	Model training & data management
GET **/api/accounts** · **/api/sessions**, POST/PATCH/DELETE variants	Ad	Account & session administration
GET **/api/stream** (SSE) · **/api/stream/feed**, POST **/api/stream/start** · **/stop**	V / Ad	Live event stream

```
# automation example: pull CRITICAL alerts into your SIEM job
curl -s -H "X-API-Key: $ATHENA_API_KEY" \
  "https://athena.example.com/api/alerts?min_level=CRITICAL" | jq .
```

12. Reports, Exports & Compliance Evidence

```
athena report           # timestamped HTML + JSON into <data-home>/reports/
athena mitre            # ATT&CK coverage + live technique hits
athena compliance       # framework control coverage
athena feedback-stats   # per-detector precision
```

- **HTML report** — styled executive/SOC report: KPIs, top risks, full alert detail with explanations. Print to PDF from any browser.
- **JSON report** — the same data machine-readable, for pipelines.
- **CSV export** — **GET /api/export.csv** or the Alerts view: one row per alert with detectors, MITRE techniques and top factor.
- **Compliance view** — every framework control lists the detectors covering it and the number of live alert evidences; ideal for audit walkthroughs.

13. Operations & Maintenance

13.1 Environment variables

Variable	Purpose
ATHENA_HOME	Data directory (DB, model, reports, config)
ATHENA_AUTH_ENABLED	Require sign-in (default **true**)
ATHENA_ADMIN_USER / **ATHENA_ADMIN_PASSWORD**	First-admin bootstrap; without the password a one-time value is generated and logged once
ATHENA_ALLOW_SIGNUP / **ATHENA_SIGNUP_ROLE**	Self-service registration toggle & granted role (never admin)
ATHENA_API_KEY	Machine principal for **X-API-Key**
ATHENA_ORG	Organization name shown in console & reports
ATHENA_SPLUNK_HEC_URL/TOKEN, **ATHENA_ELASTIC_***, **ATHENA_SLACK_WEBHOOK**, **ATHENA_SYSLOG_HOST**, …	Integration credentials (see chapter 9)

13.2 Backup, restore, retention

```
athena backup ~/backups/athena-$(date +%F).sqlite3   # safe while serving
athena restore ~/backups/athena-2026-07-20.sqlite3   # asks for confirmation
athena purge          # apply retention policy (event/alert/audit day limits)
athena reset          # wipe corpus (keeps accounts, audit; --full adds cases)
```

13.3 Monitoring

- **GET /api/health** — liveness (no DB touch); **GET /api/ready** — readiness (DB check). Wire these into your orchestrator.
- **GET /api/metrics** — Prometheus text format: request counts, 5xx, auth failures, throttles, rate limits, event/alert gauges, per-route latency.

- **athena doctor** — one-shot deployment grade: data-home writability, DB integrity, model state, auth posture, enforcement gate, retention.
- **athena monitor** — live terminal feed of the event stream with scores.

13.4 Troubleshooting

Symptom	Fix
401 authentication required	Sign in, or send **X-API-Key**; check **athena users list**
403 missing or invalid CSRF token	Send the **athena_csrf** cookie value in **X-CSRF-Token** on mutations
429 too many attempts	Login throttle — wait the **Retry-After** seconds
Lost admin password	**athena users passwd admin** on the host (CLI has data-home access)
no events; generate, ingest or load a dataset first	The corpus is empty — chapter 4
Rows skipped on ingest	Check the three required CSV columns and timestamp format
Console unreachable remotely	**--host 0.0.0.0**, open the port, prefer a TLS proxy

14. Glossary

Term	Meaning
UEBA	User & Entity Behavior Analytics — baselining normal behavior to find deviations
Risk score	Fused 0–100 score per user per analysis window
Detector	One named detection capability (rule, ML, peer or sequence based)
Dedup key	Stable alert identity (user + window) that survives re-analysis
Watchlist multiplier	Score amplifier (1–3×) for users under heightened scrutiny
Contamination	Isolation-Forest parameter: expected fraction of anomalous samples
MITRE ATT&CK	Industry framework of adversary techniques (mapped per detector)
CEF / LEEF	SIEM wire formats emitted by the syslog forwarder
SSE	Server-Sent Events — the live stream transport used by the console

— End of the ATHENA ITD User Guide —